# GLOBAL MEAN EC-Earth4 n006 1991 1999

| | ALL Global | ALL North Midlat | ALL Tropical | ALL South Midlat | DJF Global | DJF North Midlat | DJF Tropical | DJF South Midlat | JJA Global | JJA North Midlat | JJA Tropical | JJA South Midlat |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2m Temperature (land-only) [celsius] | 13.45<br>*13.70* | 2.43<br>*2.65* | 23.78<br>*24.24* | 14.72<br>*14.66* | 5.25<br>*6.53* | -13.10<br>*-11.46* | 21.50<br>*22.70* | 21.32<br>*20.51* | 20.87<br>*20.19* | 17.58<br>*16.59* | 24.85<br>*24.60* | 7.84<br>*8.66* |
| Mean Sea Level Pressure [hPa] | 1011.14<br>*1011.43* | 1016.35<br>*1014.76* | 1012.00<br>*1013.20* | 1004.13<br>*1004.66* | 1011.45<br>*1011.58* | 1018.07<br>*1015.89* | 1011.35<br>*1012.99* | 1005.02<br>*1004.55* | 1010.83<br>*1011.30* | 1013.64<br>*1012.86* | 1013.03<br>*1013.66* | 1003.41<br>*1005.09* |
| Precipitation [mm/day] | 2.70<br>*2.85* | 1.92<br>*2.15* | 3.36<br>*3.40* | 2.10<br>*2.45* | 2.69<br>*2.80* | 2.05<br>*2.27* | 3.46<br>*3.44* | 1.71<br>*2.06* | 2.77<br>*2.93* | 1.89<br>*2.12* | 3.34<br>*3.40* | 2.46<br>*2.80* |
| Evaporation [mm/day] | -2.70<br>*-2.91* | -1.47<br>*-1.74* | -3.77<br>*-3.95* | -1.71<br>*-1.99* | -2.70<br>*-2.87* | -1.53<br>*-1.76* | -3.86<br>*-3.99* | -1.44<br>*-1.74* | -2.79<br>*-3.01* | -1.45<br>*-1.76* | -3.83<br>*-4.03* | -1.94<br>*-2.20* |
| Precip. minus Evap. [mm/day] | -0.01<br>*0.02* | 0.45<br>*0.53* | -0.42<br>*-0.47* | 0.38<br>*0.50* | -0.01<br>*0.04* | 0.52<br>*0.61* | -0.40<br>*-0.45* | 0.27<br>*0.42* | -0.02<br>*-0.01* | 0.43<br>*0.51* | -0.49<br>*-0.56* | 0.52<br>*0.57* |
| Total Cloud Cover [frac] | 0.65<br>*0.68* | 0.71<br>*0.68* | 0.57<br>*0.62* | 0.75<br>*0.79* | 0.65<br>*0.67* | 0.75<br>*0.69* | 0.58<br>*0.62* | 0.71<br>*0.77* | 0.65<br>*0.67* | 0.67<br>*0.66* | 0.57<br>*0.62* | 0.78<br>*0.79* |
| Low Cloud Cover [frac] | 0.34<br>*0.38* | 0.43<br>*0.44* | 0.22<br>*0.27* | 0.49<br>*0.56* | 0.35<br>*0.39* | 0.47<br>*0.49* | 0.22<br>*0.26* | 0.51<br>*0.53* | 0.34<br>*0.38* | 0.44<br>*0.40* | 0.23<br>*0.27* | 0.48<br>*0.57* |
| Medium Cloud Cover [frac] | 0.32<br>*0.25* | 0.41<br>*0.32* | 0.20<br>*0.16* | 0.48<br>*0.35* | 0.33<br>*0.26* | 0.49<br>*0.38* | 0.21<br>*0.17* | 0.42<br>*0.31* | 0.31<br>*0.24* | 0.31<br>*0.26* | 0.20<br>*0.17* | 0.52<br>*0.38* |
| High Cloud Cover [frac] | 0.41<br>*0.34* | 0.42<br>*0.34* | 0.40<br>*0.34* | 0.44<br>*0.34* | 0.41<br>*0.34* | 0.45<br>*0.34* | 0.41<br>*0.35* | 0.38<br>*0.31* | 0.41<br>*0.34* | 0.37<br>*0.33* | 0.38<br>*0.33* | 0.51<br>*0.36* |
| Precipitation (ocean) [Sv] | 11.92<br>*13.39* | 1.71<br>*2.10* | 7.33<br>*7.86* | 2.88<br>*3.44* | 12.01<br>*13.31* | 2.23<br>*2.56* | 7.46<br>*7.88* | 2.32<br>*2.88* | 12.17<br>*13.48* | 1.22<br>*1.67* | 7.54<br>*7.88* | 3.41<br>*3.93* |
| Precip. minus Evap. (ocean) [Sv] | -1.80<br>*-1.28* | 0.26<br>*0.33* | -2.54<br>*-2.25* | 0.48<br>*0.62* | -2.14<br>*-1.46* | 0.16<br>*0.18* | -2.64<br>*-2.24* | 0.34<br>*0.57* | -1.49<br>*-1.22* | 0.48<br>*0.66* | -2.63<br>*-2.58* | 0.66<br>*0.68* |
| Precipitation (land) [Sv] | 4.00<br>*3.44* | 1.07<br>*1.07* | 2.79<br>*2.19* | 0.15<br>*0.19* | 3.86<br>*3.25* | 0.74<br>*0.80* | 2.97<br>*2.29* | 0.15<br>*0.16* | 4.17<br>*3.81* | 1.51<br>*1.45* | 2.53<br>*2.16* | 0.13<br>*0.20* |
| Precip. minus Evap. (land) [Sv] | 1.76<br>*1.41* | 0.40<br>*0.46* | 1.29<br>*0.84* | 0.07<br>*0.11* | 2.07<br>*1.70* | 0.60<br>*0.73* | 1.43<br>*0.92* | 0.04<br>*0.05* | 1.39<br>*1.18* | 0.15<br>*0.09* | 1.15<br>*0.93* | 0.10<br>*0.17* |
| TOA Net [W/m2] | 5.75<br>*0.90* | -38.36<br>*-43.38* | 45.49<br>*45.03* | -33.08<br>*-43.10* | 12.81<br>*7.92* | -120.01<br>*-128.23* | 49.87<br>*50.40* | 68.29<br>*59.10* | -3.60<br>*-7.50* | 48.60<br>*46.85* | 28.16<br>*27.81* | -122.12<br>*-132.48* |
| TOA SW Net [W/m2] | 238.31<br>*241.34* | 174.80<br>*180.65* | 301.01<br>*305.53* | 170.94<br>*173.64* | 242.93<br>*245.58* | 71.91<br>*74.74* | 303.53<br>*309.38* | 287.43<br>*288.82* | 232.11<br>*236.51* | 285.81<br>*293.85* | 284.54<br>*289.48* | 68.94<br>*73.23* |
| TOA LW Net [W/m2] | -232.56<br>*-240.44* | -213.16<br>*-224.03* | -255.52<br>*-260.50* | -204.02<br>*-216.75* | -230.12<br>*-237.66* | -191.92<br>*-202.97* | -253.67<br>*-258.98* | -219.14<br>*-229.73* | -235.71<br>*-244.01* | -237.21<br>*-247.00* | -256.38<br>*-261.67* | -191.06<br>*-205.71* |
| TOA SW Net (clear sky) [W/m2] | 285.29<br>*287.05* | 218.17<br>*224.07* | 346.61<br>*348.53* | 224.39<br>*227.04* | 293.51<br>*295.12* | 90.55<br>*95.21* | 351.86<br>*353.90* | 374.65<br>*377.49* | 278.77<br>*280.40* | 359.93<br>*363.27* | 329.27<br>*331.56* | 92.20<br>*95.19* |
| TOA LW Net (clear sky) [W/m2] | -258.93<br>*-268.35* | -238.11<br>*-248.53* | -281.83<br>*-290.60* | -231.93<br>*-243.67* | -255.77<br>*-265.44* | -215.87<br>*-226.83* | -280.83<br>*-290.10* | -243.35<br>*-254.74* | -262.60<br>*-271.79* | -261.95<br>*-271.79* | -282.14<br>*-290.62* | -222.45<br>*-234.14* |
| SW Cloud Forcing [W/m2] | -46.97<br>*-45.70* | -43.37<br>*-43.42* | -45.60<br>*-43.00* | -53.45<br>*-53.40* | -50.58<br>*-49.54* | -18.64<br>*-20.46* | -48.32<br>*-44.52* | -87.22<br>*-88.67* | -46.67<br>*-43.88* | -74.11<br>*-69.42* | -44.73<br>*-42.08* | -23.26<br>*-21.95* |
| LW Cloud Forcing [W/m2] | 26.37<br>*27.91* | 24.94<br>*24.50* | 26.31<br>*30.10* | 27.91<br>*26.92* | 25.65<br>*27.78* | 23.95<br>*23.86* | 27.16<br>*31.12* | 24.21<br>*25.01* | 26.89<br>*27.78* | 24.74<br>*24.79* | 25.76<br>*28.95* | 31.39<br>*28.43* |
| Surface Net SW [W/m2] | 160.19<br>*160.00* | 110.06 | 207.32 | 111.91 | 164.74 | 44.48 | 210.01 | 190.48 | 153.30 | 179.35 | 193.35 | 43.62 |
| Surface Net LW [W/m2] | -58.19<br>*-56.00* | -54.55 | -63.59 | -50.58 | -57.34 | -52.09 | -63.60 | -49.52 | -57.76 | -53.15 | -62.85 | -51.73 |
| Surface SH [W/m2] | -17.59<br>*-21.00* | -19.69 | -20.45 | -9.52 | -16.58 | -19.06 | -20.00 | -6.95 | -19.21 | -22.35 | -20.91 | -12.53 |
| Surface LH [W/m2] | -78.32<br>*-82.00* | -42.67 | -109.15 | -49.61 | -78.23 | -44.44 | -111.82 | -41.89 | -80.64 | -42.13 | -110.89 | -56.00 |
| Net Surface (no snowfall) [W/m2] | 6.08<br>*0.60* | -6.85 | 14.14 | 2.20 | 12.59 | -71.11 | 14.59 | 92.12 | -4.31 | 61.71 | -1.30 | -76.63 |
| Net Surface [W/m2] | 5.33<br>*0.60* | -8.23 | 14.13 | 0.52 | 11.74 | -73.65 | 14.56 | 91.24 | -4.93 | 61.46 | -1.30 | -78.92 |
| TOA - Sfc Net Radiation (no snow) [W/m2] | -0.33<br>*0.00* | -31.52 | 31.35 | -35.28 | 0.22 | -48.91 | 35.28 | -23.83 | 0.71 | -13.12 | 29.46 | -45.49 |
| TOA - Sfc Net Radiation [W/m2] | 0.43<br>*0.00* | -30.13 | 31.37 | -33.60 | 1.07 | -46.36 | 35.30 | -22.95 | 1.33 | -12.87 | 29.46 | -43.20 |

Model Bias
(standard deviation of interannual variability from observations)
10
9
8
7
6
5
4
3
2
1
0
−1
−2
−3
−4
−5
−6
−7
−8
−9
−10